

Introduction

This project analyzes daily screen time behavior using real personal data collected from mobile devices. The aim is to understand how screen usage changes under different contexts, such as exam days, activity days, and weekends, and to evaluate whether daily screen time can be predicted using basic machine learning models.

Variable	Description	Type
date	Observation date	Date
total_minutes	Total screen time in minutes	Numerical
social_minutes	Time spent on social apps	Numerical
productivity&finance_minutes	Time spent on productivity or finance apps	Numerical
entertainment_minutes	Time spent on entertainment apps	Numerical
creativty_minutes	Time spent on creativity apps	Numerical
travel_minutes	Time spent on travel apps	Numerical
shopping&food_minutes	Time spent on shopping or food apps	Numerical
reading_minutes	Time spent on reading apps	Numerical
games_minutes	Time spent on game apps	Numerical
other_minutes	Time spent on apps which belong to other areas	Numerical
notifications	Number of notifications received	Numerical
pickups	Number of device pickups	Numerical
exam day	1 if exam day, else 0	Binary
activity	1 if exercise/social activity day, else 0	Binary
weekend	1 if weekend, 0 otherwise	Binary

Figure #1

Data

In *Figure #1* you can see how screen time data is formatted into csv file. Two csv files named `myScreenTime.csv` and `myScreenTime_2.csv` are used along the project. The first one is used for EDA & hypothesis testing part and also for the ML model training, while the second file was used to compare findings of the ML models.

Exploratory Data Analysis (EDA) Results

Results of exploratory analysis of data on overall screen time indicated high variability in overall screen time daily, where productivity and social applications dominated overall daily screen time. However, exploration of overall screen time through other categories such as creativity and reading indicated low activity. Analysis of overall screen time in relation to notifications/device pickup indicated a weak positive association. Additionally, overall screen time was lower on exam/activity days than on regular days. You can see the related graphs in the notebooks.

Hypothesis Testing Results

Productivity-related screen time was analyzed across three exam-related periods (*Before Exam*, *Exam Day*, and *After Exam*) based on the hypothesis that academic workload increases before exams and decreases once the exam is taken, particularly because productivity apps such as note-taking tools are used more intensively during exam preparation. Accordingly, the null hypothesis stated that there is no difference in productivity screen time between these periods, while the alternative hypothesis predicted higher productivity usage before exams and lower usage during and after the exam day. A one-way ANOVA revealed a statistically significant difference between groups ($F = 8.56$, $p = 0.0007$), indicating that productivity screen time varies across exam periods. Post-hoc Tukey HSD tests showed that productivity usage was significantly higher before exams compared to exam days (mean difference ≈ 297 minutes, $p < 0.01$) and days after exams (mean difference ≈ 256 minutes, $p < 0.01$), while no significant difference was observed between exam days and days after exams. These results support the alternative

hypothesis and indicate a clear behavioral trend in which productivity-related screen usage increases leading up to exams and decreases once the exam period begins.

Machine Learning Results

The ML results show that predicting daily screen time is a challenging task using the existing context and interaction-based features. The results for Linear Regression on all three sets of features (context, interaction, and joint) show negative R^2 values on the time-ordered holdout set, revealing that it is no better than a naive model predicting the average screen time. The context and joint models have comparable MAE and RMSE results, meaning that it is not beneficial to include interaction features for linear regression, while the result for interaction features fared considerably poorer. These observations specify that there is not a relevant linear relationship between screen time and its predicting factors.

Then, Decision Tree Regression was applied to model any potential nonlinearities and conditional relationships. While some of the decision tree models attained slightly smaller errors on the training holdout set, they strongly deteriorated on the external evaluation dataset with strongly negative R^2 values. The gap between internal to external performance suggests that models overfit although they were able to learn trading day patterns within the training period, they could not generalize to unseen future data. Overall, results indicate that daily screen time behavior is very volatile and noisy, which, along with limited data and simple models, is not sufficient for reliable prediction, even when nonlinear methods are used.

You can see the results of Linear Regression model at *Figure #2* and Decision Tree model at *Figure #3*

	Model	Dataset	MAE	RMSE	R2
0	A (Context-only)	Train Holdout	111.852886	133.976424	-0.181811
1	B (Interaction-only)	Train Holdout	149.167908	167.256703	-0.841867
2	C (Combined)	Train Holdout	112.060058	133.529920	-0.173946

Figure #2

	Model	Dataset	MAE	RMSE	R2
0	DT A (Context-only)	Train Holdout	130.27	149.54	-0.47
1	DT A (Context-only)	External df_eval	209.15	230.82	-10.73
2	DT B (Interaction-only)	Train Holdout	215.20	263.63	-3.58
3	DT B (Interaction-only)	External df_eval	163.24	204.82	-8.23
4	DT C (Combined)	Train Holdout	107.80	171.78	-0.94
5	DT C (Combined)	External df_eval	147.35	182.44	-6.33

Figure #3

Conclusion

This study examined daily screen use based on actual personal data as well as other considerations such as exam weeks, activity weeks, and weekends. Both exploratory analysis and hypothesis testing showed definitive results, particularly within the context of exams where productivity screen use rose before exams but reduced afterwards.

Nevertheless, the machine learning outcomes indicated that screen time per day cannot be predicted well. Linear Regression Models were ineffective at explaining significant relations, and Decision Tree Models resulted in overfitting, thus not being a good fit for new outcomes. In general, the outcomes indicate a complex and volatile screen time behavior, which cannot be predicted with enough accuracy using machine learning models with a small dataset.